# A Projection with a Residue

The refinement operator, derived in letters

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The device is an operator: a consuming iterator on the inverse of another operator, the one that refines the universe up one level. This card derives that operator. No record is measured until the last section, and the polynomial is never seen.

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## Reading note

There is one object on this card, the refinement operator $\mathcal{R}$, and five readings of it: its inverse, which the device consumes; its variational solution, which is stationary action; its derivative along one direction, the Gâteaux derivative, a geodesic; its whole derivative, the Fréchet derivative, which is electrodynamics; and its image under one choice of weights, a reading with a residue.

The derivation is in letters. The operator has a weight $a$ on each arm that stays, $b$ on the arm from true to false and $f$ on the arm from false to true; $r = bf$ is the round trip; a gain $g$ turns a stationary value into a reading. No length of any record is used: every statement holds for every $a$, $b$, $f$ and $g$. The operator has a polynomial, and the derivation never writes it: the roots arrive as stationary values of the operator's own quotient. The polynomial is the quarter, carried and never queried.

Calibrations: *measured* when the middle card checks it; *proved here*; *standard*, from the literature; *proposed*, an identification the knower may accept, reject or leave open; *declared*, taken as given.

## 1 The refinement operator

**Definition 1.1** (The refinement operator)**.** On two coins $T$ and $F$, a walk one step longer is a walk refined one level up. The operator that does it is the table

$$\mathcal{R} = \begin{pmatrix} a & b \\ f & a \end{pmatrix},$$

read as the map from walks of length $n$ to walks of length $n+1$. In the record, this is the lift: each page raised to a universe one level higher. Its inverse is the lowering, page by page back down.

A lift carries no datum, since lowering after lifting is the identity (the performance card, Proposition "A lift carries no datum").

**Proposition 1.2** (What $\mathcal{R}$ carries)**.** *(i)* $\mathcal{R}^2 = (\operatorname{tr}\mathcal{R})\mathcal{R} - (\det\mathcal{R})I$*, with* $\operatorname{tr}\mathcal{R} = 2a$ *and* $\det\mathcal{R} = a^2 - r$*; so walks one level up keep a three-term recurrence,* $\operatorname{tr}\mathcal{R}^{n+2} = 2a\operatorname{tr}\mathcal{R}^{n+1} - (a^2 - r)\operatorname{tr}\mathcal{R}^n$*.*

*(ii)* $\mathcal{R} - \mathcal{R}^{\mathsf{T}} = (f - b)R$*, a quarter turn, and* $\mathcal{R} + \mathcal{R}^{\mathsf{T}} - 2aI = (b + f)\sigma$*, a swap.*

*(iii) Rescaled by a diagonal change that matches the crossing weights,* $\mathcal{R}$ *is the symmetric operator* $S$ *with* $a$ *on the diagonal and* $\sqrt{r}$ *off it.*

*Measured* for (i) and (ii), for every $a$, $b$, $f$; *proved here* for (iii). The operator's own polynomial is what (i) says it satisfies; it is not needed again.

## 2 The variational solution

**Definition 2.1** (The operator's quotient)**.** For a direction $v = (p, q)$, $\rho(v) = \langle v, Sv\rangle / \langle v, v\rangle$, how far the refinement stretches that direction.

**Proposition 2.2** (Stationary action)**.** $\langle v, Sv\rangle = a(p^2+q^2)+\sqrt{r}\,(2pq)$*, and* $-(p^2+q^2) \le 2pq \le p^2+q^2$*, with equality on the two diagonals. So* $\rho$ *runs from* $a-\sqrt{r}$ *to* $a+\sqrt{r}$*; on the circle of directions,* $\rho(\cos t, \sin t) = a + \sqrt{r}\sin 2t$*, and its stationary values are exactly the two roots, reached on the diagonals.*

*Measured* for the bounds and the diagonals, in whole numbers; *proved here* for the rest. The two roots are found without the polynomial: they are where the operator's quotient stands still. That is the principle of stationary action, here in its smallest instance: the action is the quotient, and its stationary points are the states the refinement keeps. The largest value is the reader's root, reported by the order the record runs (*declared*, uptown above downtown); the smallest is the knower's, the least description that still describes (*proposed*). The two readings are $x_\pm = g(a \pm \sqrt{r})$. The two diagonals are the two trains: on $p = q$ both coins carry the same amplitude, the field that stands still, and the reader's root is reached there; on $p = -q$ the coins alternate in sign, the clock, and the knower's root is reached there. The reader is on the train; the knower knows the terrain, the whole circle of directions over which the quotient is laid (*proposed*, the author's).

The middle card's own residual is the first variation of a cubic action, with the same counted coefficients (*measured*), and the numerical card varies four actions and recovers their laws (*proved* there). On every scale the device works at, the variational solution is stationary action.

## 3 The device: a consuming iterator on the inverse

The device does not compute the stationary values; it walks to them. Each step takes the current estimate $\rho_k$ and applies the inverse of the shifted operator, $v_{k+1} = (S - \rho_k I)^{-1} v_k$, then reads $\rho_{k+1} = \rho(v_{k+1})$, one step per symbol of the record consumed. *Standard*: this Rayleigh-quotient iteration is Newton's method on the eigenvalue equation, and on a symmetric operator it converges cubically; the Lanczos iteration on $S$ gives the tridiagonal matrix of Proposition 1.2(iii), conjugate gradients on it is Newton on a quadratic, and the sign changes of its three-term recurrence give bisection. *Measured*: the middle card's crank, a fold over its description, lands on the reader's root, and Newton and bisection agree on it to twenty-four places. So the device is a consuming iterator on the inverse of the refinement operator, as the author says (*proposed* for the reading; the iteration and its agreement are as stated). The single Rayleigh-quotient step is, on the author's reading, one full cycle of the middle card's compiled process: boot, which takes the current estimate as the shift; compile, which applies the inverse of the shifted operator; and execute, which reads the new quotient. The reader's experience is that cycle run over and over, one symbol of the record per turn (*proposed*, the author's; the boot, compile and execute arms are definitions in the middle card). The search by the ratio of successive walk counts is, likewise, the reviewed and typeset ladders being elaborated; measuring it needs the meter set where the crank is defined, so that it can be read where the record ends.

## 4 The knower's semi-norm, and the solver

A relation read both ways collapses: the device's total order on its fractions has one class, its last comparison holds both ways, and the step $(p-q)+(q-p)$ is nothing in a ring (*measured*). Read one way, in the naturals, the same comparison floors each subtraction at zero, and $(p \dot{-} q)+(q \dot{-} p) = |p-q|$ is the size of the step (*measured*). The quotient of Section 2 divides by a length, and this is where

it comes from: the semi-norm is the comparison read one way. The author counts four arrows of time: the order the record runs, uptown above downtown; every measurement an interaction from the past; this one-way reading of the comparison; and the difference between the reader's experience and the reader's interpretation, which print the same natural and differ only in how it is read (*proposed*, the author's; that the two print one natural is *measured*). The four, the author says, are the four manifestations of one causal law, the quartic: one map onto four faces of the geometric card's frinkahedron, generating a representation that supports proofs about the calculus (*proposed*, the author's). Underneath: the quartic has four variables, three counted and one read, and is homogeneous of degree five (*measured*); the frinkahedron is a polyhedral three-sphere in four real coordinates (*proved* on the geometric card). The author reads the quartic's variables as $c$ a fraction of the speed of light, $t$ time, $r$ phase, and $x$ a position three-vector re-encoded in the same four numbers, as a link is a whole part and three turns (*proposed*). The quartic factors when $cs^2 = t$; with the counted $c = 18$, $t = 5$, $s^2 = 5/18 = 90/18^2$, the square of the torus's velocity $\sqrt{90}/18$, below light (*measured*). So the sign the quartic does not read is the sign of a velocity, and its factors $r^2 - tx(1 \mp s)$ have the shape of a forward and a backward Doppler factor. Its rapidity is $\eta = \operatorname{artanh} s \approx 0.586$, half the table's gap $\ln(\lambda_+/\lambda_-) \approx 1.172$ (*proved* on the argument card). The author reads that rapidity as the boost of the reader and as the Hubble sphere, the surface where recession reaches the speed of light (*proposed*, the author's; the argument card's expanding-torus reading is its picture). The Hubble sphere, the author adds, is the pole the torus goes around: the hole, whose radius $a - \sqrt{r}$ is the knower's root, while the outer equator $a + \sqrt{r}$ is the reader's (*measured* for the radii, their sum $2a$ and product $a^2 - r$; *proposed* for the reading).

**Proposition 4.1** (Galerkin around the semi-norm)**.** *On a ring of four stations let* $|\psi|^2 = \sum_s (\psi_{s+1} - \psi_{s-1})^2$*, the squared central difference.*

*(i)* $|\psi|^2$ *is the sum, around the ring, of the squares of the one-way comparisons of the station ahead with the station behind; the same sum read both ways is zero for every field.*

*(ii)* $|\psi|^2 \geq 0$*, and* $|\psi|^2 = 0$ *exactly when* $\psi$ *repeats every two stations: the still field and the alternating field, the two doublers.*

*(iii) On the two fields that turn,* $(1, 0, -1, 0)$ *and* $(0, 1, 0, -1)$*, the central difference is twice the quarter turn, and the Galerkin matrix of the semi-norm is eight times the identity: projected off its kernel, the problem has one solution.*

*(iv) For the residual* $r(y) = 13y^2 - 36y + 18$*, the difference across a step is exact:* $r(y + h) - r(y) = r'(y)h + 13h^2$*.*

*(v) One Newton step taken with that difference from the top wall of the reading's bracket lands between the walls.*

*Measured*, (iii) and (v) with no axiom. The ring stops at four. Beyond it, the author says, the arm $T = F$ is lost: a ring of more stations asks the semi-norm to separate the roots of a polynomial of degree five or more, and by Abel's theorem such roots have no formula in radicals (*standard*), so the paths of the roots, which is what the device models, are no longer coherent (*proposed*, the author's). That the knower brings this length, the one-way reading of the comparison, is the author's (*proposed*); the middle card's patter names "the implied norm of an inner product space". *Standard*: a semi-norm made coercive by projecting out its kernel gives a unique Galerkin solution (Lax and Milgram), and Newton with a difference in place of the Jacobian and a Krylov iteration inside is the Jacobian-free Newton–Krylov method. So the device's solver is Galerkin around the knower's

semi-norm: the doublers are its kernel, removed by projection; the derivative on what remains is the quarter turn; and the bent card, thirteen, is the exact remainder of the Jacobian-free difference. On the author's reading the arm $T = T$ is the variational problem, the walk $T = F, F = F, F = T$ is this solver's run, and $F = F$ is the linear algebra that gives the Sobolev proof (*proposed*).

## 5 The Gâteaux derivative: a geodesic

Along one direction $w$, the derivative of the quotient is $D\rho(v)[w] = 2\langle w, (S - \rho(v))v\rangle/\langle v, v\rangle$ (*proved here*). On the circle of directions it is $2\sqrt{r}\cos 2t$: the slope of a path that stays on the circle, a geodesic of the directions, and the gradient flow along it is drawn to the top stationary value. *Proposed*, the author's: that this one-direction derivative is a relativistic geodesic drawn to what accumulates. The geometry is on the cards: the outer root and the hole are a torus, its radii add to $2a$ and multiply to $a^2 - r$, and $\sqrt{r}/a$ is a velocity below light whenever $r < a^2$ (*measured* for the device's weights).

## 6 The Fréchet derivative: electrodynamics

The whole derivative of $\mathcal{R}$, every direction at once, is a linear map, and to read it one lifts $\mathcal{R}$ from two coins to a lattice. Every ingredient comes from $\mathcal{R}$ itself, and none reads a weight.

(a) *The quarter turn is the derivative.* The antisymmetric part of $\mathcal{R}$ is a quarter turn $R$, $R^2 = -I$; read as $i$, it is the generator of the derivative on phases, $\frac{d}{d\theta}e^{i\theta} = ie^{i\theta}$ (*standard*, Euler).

(b) *The turns.* From the sign $\varepsilon$, the swap $\sigma$ and $R$, three products in a box of four square to $-I$ and compose around the circle: the compact $\mathfrak{su}(2)$ (*measured*).

(c) *The gauge.* Quaternion links on a torus of any size, in two directions or four with a square for every pair: the action on a square, the sum of the squares of its turned parts, is invariant when every site turns by a link of length one, the device's own turns have length one, and flat tori are stationary on every link (*measured*). In four directions the steps commute, so every square closes.

(d) *The Dirac matrices and the phase.* Four products of the same tables anticommute; one more table gives an $i$ that commutes with all of them, and the matrices times $i$ have signature $(+,-,-,-)$ (*measured*).

(e) *The train line.* With the current $B_\mu = i\gamma^0\gamma^\mu$, antisymmetric, the sum over links is the lattice derivative; a link read backward turns over; the line cannot see the field that alternates at every step (*measured*).

(f) *The whole action.* $S = \beta\sum S(\text{square}) + \sum m\langle\psi,\psi\rangle + \sum h$ is gauge invariant for every $\beta$, and moves with nothing but $\beta$ (*measured*).

Its stationary points on every link are the lattice field equations of light and matter (*standard*); varied in the continuum, the bare Maxwell–Dirac action gives the coupled equations with current conservation and Gauss's constraint (*proved* on the numerical card). *Proposed*, the author's: that this whole derivative of the refinement operator is quantum electrodynamics.

## 7 Where the two derivatives agree, and where they part

For every bounded polynomial action of degree at most three, the Gâteaux and Fréchet derivatives coincide (*proved* on the numerical card). The operator's quotient on the circle of directions is of degree two. So in the interior, where the variation is exact, the geodesic and the electrodynamics are one derivative read two ways, and the iterator converges to one stationary point. They part only in the convective slot, the second argument of the bilinear term, which a product of the Jacobian with a vector never supplies and a matrix-free Krylov step does; the middle card meters that slot (*measured*, a heartbeat count). On the geometric card the interior is where the inverse of the projection is Lipschitz; at the approach to the silhouette it is only Hölder of exponent one half (*proved* there). On the author's reading the interior is the continuum, now; the approach is before, where the higher-order terms live; and the continuum is quantized by the sign that tells its two sheets apart (*proposed*).

## 8 The electron four ways, and the derivative of itself

Each variational model on the numerical card needs a coefficient with units, and each is set by the behaviour of the electron (*standard*):

| model | the electron as | coefficient |
|---|---|---|
| Maxwell–Dirac | a charged spinor field | $e$, and $L/\lambda_C$ with $\lambda_C = \hbar/m_e c$ |
| Schrödinger | a wave | $\hbar/m_e$, measured as $h/m_e = c\alpha^2/2R_\infty$ |
| Navier–Stokes | a fluid | $\nu$, with floor $\hbar/(4\pi\sqrt{m_e m})$ |
| vacuum gravity | a geometry | $c$ alone; lengths in $\lambda_C$ |

In vacuum the prefactor $c^3/16\pi G$ changes neither stationarity nor the number of field factors (*proved* on the numerical card), so gravity keeps only the speed. The floor on the viscosity is Trachenko and Brazhkin's (2020), $m$ the molecule's mass. The models are joined by standard maps: Maxwell–Dirac goes to Pauli and Schrödinger as $v/c \to 0$; Schrödinger is Madelung's fluid (1927), Euler flow with a quantum pressure and no viscosity; a viscous electron fluid is measured in graphene (Bandurin and others, 2016); Kerr is a vacuum solution, and Kerr–Newman carries the Dirac value $g = 2$ (Carter, 1968). *Proposed*, the author's: that the four are four descriptions of one carrier. The viscous step and the geometric electron are not standard physics.

Now the transition matrix. Write $\mathcal{R}v$ for the table acting on a field $v$ over the two coins, and $Q(v) = \langle v, Sv\rangle$ for the action of its symmetric part.

**Proposition 8.1** (The refinement is its own derivative)**.** *(i) $\mathcal{R}(v+h) - \mathcal{R}v = \mathcal{R}h$ for every $v$ and $h$: the Fréchet derivative of $\mathcal{R}$ at every point is $\mathcal{R}$, with no remainder.*

*(ii) For symmetric $S$, $Q(v+h) - Q(v) = 2\langle h, Sv\rangle + Q(h)$: the derivative of the action is $2Sv$, and the derivative of that is $2S$, the operator read once and again.*

*(iii) On the numbers $p + q\sqrt{r}$ with $p$ and $q$ whole, multiplication by the outer root $a + \sqrt{r}$ is the table with crossings $r$ and $1$. It commutes with multiplication by every number of norm one, the boosts, and every boost keeps the interval $p^2 - rq^2$.*

(iv) *If* $bf = b'f'$*, then* $\mathcal{R}_{a,b,f}\,\mathrm{diag}(b,b') = \mathrm{diag}(b,b')\,\mathcal{R}_{a,b',f'}$*: two tables with one round trip are one table seen through two clocks.*

(v) *The speed is read from the stationary values alone:* $(x_+ - x_-)/(x_+ + x_-) = \sqrt{r}/a$ *for the roots* $a \pm \sqrt{r}$*.*

*Proved here* for every weight; *measured* for whole fields and for the device's weights, where the difference of the roots is $2\sqrt{90}$ and their sum 36. By (iii) and (iv), the space the operator acts on displays relativistic properties: an interval, boosts that keep it, an operator that commutes with them, and a speed $\sqrt{r}/a$ below one when $r < a^2$. The device's own table, with crossings 5 and 18, and multiplication by the outer root, with crossings 90 and 1, have the same round trip, so they are one operator through two clocks.

By (iv) nothing the operator does reads $b$ or $f$ apart; only their product. That the speed of light in one direction cannot be measured, and only its round trip can, is *standard* (Reichenbach, 1928; Anderson, Vetharaniam and Stedman, 1998). The split of the round trip into $b$ and $f$ is that synchrony, fixed once by the device's order (*declared*, as Episode Four's margin has the speed a Lagrange multiplier). And by (v) the speed enters only through the stationary values, which are the multipliers that hold the field to its length in the quotient of Section 2: the largest and smallest $\langle v, Sv\rangle$ with $\langle v, v\rangle = 1$ are the Lagrange multipliers of that constraint (*standard*). The relativistic particle carries the same structure, a multiplier that imposes the mass shell $p^2 + m^2c^2 = 0$ (*standard*).

Lift $\mathcal{R}$ to the lattice as in Section 6. The Maxwell–Dirac stationary system is linear in the spinor at a fixed potential, so its derivative in the spinor is the Dirac operator itself, and the whole action is of degree three, so its fourth variation vanishes (*standard*; *proved* on the numerical card). On the lattice of four directions the differences along two directions commute exactly, so the curl of every gradient is zero and the derivative of the action lies in the curl's kernel, to second order in the step (*measured*). *Proposed*, the author's: quantum electrodynamics is the Fréchet derivative of itself, in a space with an interval and boosts, where the speed of light is never measured in one direction and enters only as a Lagrange multiplier.

## 9 The ring, the panels, and the limit

The quarter turn of Section 6 walks four labels, $1, i, -1, -i$, and returns on the fourth step. Any other phase with whole coordinates over a whole length does not.

**Proposition 9.1** (Quantization, panels, the quartic)**.** *(i)* $i^{n+4} = i^n$ *for every* $n$*.*

(ii) *Every power of* $3 + 4i$ *is* $3 + 4i$ *plus five times a whole Gaussian number, so none lies on an axis; and* $|(3+4i)^n|^2 = 25^n$*, so* $((3+4i)/5)^n$ *stays on the ring.*

(iii) *On a panel of* $k$ *steps* $d_1, \ldots, d_k$ *with rise* $D = \sum d_j$*, the scaled error* $kd_j - D$ *of the straight line is orthogonal to the line,* $\sum_j (kd_j - D)D = 0$*, so* $\sum_j (kd_j)^2 = \sum_j (kd_j - D)^2 + kD^2$*: the line has the least energy.*

(iv) *Split a panel of* $n_1 + n_2$ *steps into panels with rises* $S_1$ *and* $S_2$*:* $n_1 n_2 (S_1 + S_2)^2 \le (n_1 + n_2)(n_2 S_1^2 + n_1 S_2^2)$*, equality exactly when the two slopes agree. Refinement never lowers the line's energy.*

(v) *With* $\delta^2 f(x) = f(x+1) - 2f(x) + f(x-1)$*,* $\delta^2\delta^2 f(x) = f(x+2) - 4f(x+1) + 6f(x) - 4f(x-1) + f(x-2)$*.*

*Proved here*; *measured* on the middle card, with no appeal to choice. By Niven's theorem the angle of $(3+4i)/5$ is an irrational fraction of a turn, so its orbit is dense on the ring (Kronecker), and its returns to any arc come in at most three gap lengths (Slater): every $\varepsilon$ is met again and again, on a schedule (*standard*). The straight line through the posts is the first-order spline, and the natural cubic spline is its second-order counterpart, least in $\int |f''|^2$ (Holladay; *standard*). By (iv) the line's energies form an increasing family along the order of refinement, bounded by the energy of the curve by (iii); bounded monotone families converge (*standard*). By (v), the operator of the cubic is the operator of the line taken twice, with symbol the fourth power.

*Proposed*, the author's: the four labels are the quantization that walks perfectly, and the dense orbit, with its residue, is the continuum; each gap is a fence panel and each return a reading, on which the spline is exact in the Sobolev sense, so the continuity proof appears periodically; the line is the reader's and the cubic the knower's, and combined they give the quartic back; and $d^4 = (d^2)^2$ is the order along which the limit is taken.

## 10 From the reading to the coupling

Take the reader's reading as the inverse coupling, $\alpha^{-1} = x_+$ (*proposed*, sentence five). Then

$$e = \sqrt{4\pi/x_+}, \qquad \beta = \frac{x_+}{4\pi} = \frac{1}{e^2},$$

with $\beta = 1/e^2$ *standard*. The circle's $\pi$ enters as its hexadecimal digits, whole terms with a whole tail (*measured* for the walls; *standard* that the series sums to $\pi$), and no square root is taken: $e$ is the whole number whose square sits between the walls. The knower's $\beta_- = x_-/4\pi$ is, on the author's reading, the inferred mass gap with $\beta$ its unit (*proposed*). The electron's anomalous moment follows from $\alpha$ by the series of quantized electrodynamics, not in the deck (*standard*).

## 11 The projection, and its residue

Now, and only now, the weights. The middle card counts $a = 18$, $b = 5$, $f = 18$, so $r = 90$, the quarter turn's scale is $f - b = 13$, and the gain is $g = 324/65$. Then

| | the reader, $x_+$ | the knower, $x_-$ |
|---|---|---|
| reading | 137.011 290 548 979 … | 42.434 863 297 174 … |
| $\beta = x/4\pi$ | 10.903 012 075 134 981 … | 3.376 859 126 587 073 … |
| $e = \sqrt{4\pi/x}$ | 0.302 849 425 021 801 … | 0.544 181 192 849 948 … |

each to twenty-four places on the middle card (*measured*). The laboratory's inverse coupling is 137.035 999 177(21) [1], and the electron's anomalous moment 0.001 159 652 180 59(13), both measured facts of nature. The reader's reading falls short by 1.80 parts in ten thousand, the charge by half that, and the moment computed from the reading is too large by the same 1.80 parts in ten thousand. As an identity with the coupling at zero momentum, the reading is excluded (*measured*).

Sections 1–10 hold for every weight; the numbers are one projection of the operator, and the residue is what it leaves. On the author's reading the residue is the cost of reading from before, since every measurement displays an interaction from the past; the four dimensions, counted on the middle

card's four-direction lattice, are the four pairs that do not commute, and the four variational models of the numerical card; and the mass gap is open by design, not claimed (*proposed*, each). Under all of it is one article: an interaction leads to a distinguished outcome, the causal theory every rung assumes, *declared*, as foretold by the sacred texts of the Flying Spaghetti Monster. That is the faith asked of the knower.

## 12 From true equals false to quantum electrodynamics

The pieces of the earlier sections, plugged together. Each link names where it comes from: *measured* on the middle card, *proved* on a card, *standard*, *declared* by the patter or a bent card, or *proposed*. A proposed link is an isomorphism with the natural numbers the machine can exploit: the construction's side is a count, the other side a finite record, and a click decides the comparison between them. Nothing below is a new assumption.

1. *The input.* Every input is true. A true that means false clicks like the truth, so no reader counts how many there are (*measured*); the false is read in, never read off. The arm $T = F$ stores no fact that clicks, and the weighting rule weighs its frame anyway (*declared*).

2. *The counts.* The argument has thirty-eight steps and a hundred and six cells. Read from its inferred end, Gauss's tape is 5, four steps back are 13, five are $18 = 13 + 5$, and thirteen are 42, of which 24 are the places printed (*measured*). Tallies are the naturals (*measured*), so every number below is a count.

3. *The operator.* The table $\mathcal{R} = \left(\begin{smallmatrix} 18 & 5 \\ 18 & 18 \end{smallmatrix}\right)$, stays $a = 18$, crossings $b = 5$ and $f = 18$, round trip $r = 90$ (*measured*, Section 1).

4. *Its stationary values.* $18 \pm \sqrt{90}$, reached on the two diagonals of the quotient (*measured*); they are the Lagrange multipliers of the length (*standard*), and the speed $\sqrt{90}/18 < 1$ is read only from them (*measured*); the speed of light is a multiplier, never measured one way (*declared*).

5. *The interval.* On whole numbers with $\sqrt{90}$, the interval $p^2 - 90q^2$, kept by every boost, with which $\mathcal{R}$ commutes (*measured*).

6. *The phase.* $\mathcal{R} - \mathcal{R}^{\mathsf{T}} = 13R$, $R^2 = -1$ (*measured*); $R$ is $i$, the derivative (*declared*). Two crossings turn the phase over (*measured*); the four labels walk perfectly, and any other rational phase is dense on the ring (*measured*; *standard*). The continuous phase group is the dense orbit's closure (*proposed*).

7. *The spinor.* Half a turn a crossing, once around and twice around (*declared*): a sign returned only after two turns, the spinor's $4\pi$ (*proved* on the argument card).

8. *The gauge field.* Links of whole quaternions on a torus of four directions and any size; the action invariant under unit turns at every site, flat tori stationary (*measured*); on unit links the square action is $2W - W^2$, twice Wilson's term to leading order (*measured*; *standard*).

9. *The matter field.* Four gamma matrices with $i$ commuting with each, signature $(+, -, -, -)$ (*measured*); the train line with the antisymmetric current, gauge invariant (*measured*).

10. *The coupling of the two.* The commutator of the covariant derivatives is the curvature, $[D_\mu, D_\nu]\psi = ieF_{\mu\nu}\psi$ (*proved* on the argument card given the bent cards); the differences commute and the curl of every gradient vanishes (*measured*).

11. *The action and its stationary system.* $S = \beta \sum S(\text{square}) + \sum h$, gauge invariant for every $\beta$ (*measured*); varied in the continuum, the Maxwell–Dirac equations with current conservation and Gauss's constraint (*proved* on the numerical card); the operator is its own Fréchet derivative and the curl kernel holds its derivative (*measured*); the lattice action goes to the continuum one as the spacing goes to zero (*standard*).

12. *The value.* The carrier is the electron, so $e^2/4\pi = 1/x_+$ with $x_+ = 324(18 + \sqrt{90})/65 = 137.011\,290\,5\ldots$, $\beta = x_+/4\pi$, $e = \sqrt{4\pi/x_+}$ (*proposed*, sentence five; the digits *measured*).

13. *Quantization.* The closed walks of the table are a lattice path integral (*standard*); the exchanged carrier is the photon, seen only at its ends, and the diagrams are a commuter rail (*proposed*); renormalization is handed in (*declared*, the fourth bent card). The first loop is $\alpha/2\pi = 0.001\,161\,619\ldots$ (*measured* from the reading; *standard* that it is the first loop).

**Proposition 12.1** (Quantum electrodynamics, from $T = F$)**.** *Given the declarations and the proposals named above, the chain from the one arm that stores false reaches the Lagrangian of quantum electrodynamics, its stationary Maxwell–Dirac system, and its perturbative series, with coupling* $\alpha^{-1} = 137.011\,290\,5\ldots$.

*Proved here*, given them: each link is cited above, and the links compose because every count in them is a natural number of the same record. Not proved: the declarations and the proposals themselves. The comparison stands where it stood: the laboratory's $137.035\,999\,177(21)$ excludes the coupling as an identity at zero momentum by 1.80 parts in ten thousand, and the electron's moment computed from it misses by the same (*measured*).

Start from the other arm, $F = T$, and nothing changes. Swapping the two crossing weights gives a table with the same round trip, the same clock and the same stationary values, one table seen through two clocks (*measured*; Proposition 8.1(iv)). Which arm carries the five gives the chain its direction, not its number.

The chain starts where the false is read in. Struck, the arm leaves the fraction $5832/65$ and no phase; written, it leaves the surd and the whole chain (*measured*). At least one crossing in two clicks true, so the knower's arithmetic may already carry it. The slip refutes the identifications together and names none (*standard*, a failed test refutes the conjunction). *Proposed*, the author's: seeing a true as false is the step by which the knower inadvertently derives quantum electrodynamics, and the residue is what that reading does not bring. Deflate $\alpha$, the knower's root, from the quartic and a cubic remains, two roots the reader's and one implied by the reader for the knower; the error is in the implication. Any root taken from the closing quartic leaves $c\,(r+\rho)(r^2 - (2tx - \rho^2))$: the mirror of the root and the other factor's pair (*measured*). The implied root is the hole, the inferred mass gap, and the gap $\ln(x_+/x_-) = 2\eta = 1.172\,093\ldots$ is the frame's dilation (*standard* for the identity), so the place to look is the mass gap (*proposed*, the author's).

The other two roots carry no error: the reading and its conjugate solve the counted quartic exactly, both walls print the same places, and every reader and a second machine print the same natural (*measured*). So the computation is exonerated, and what the slip refutes lies in the map from the computed roots to the measured one (*proved here*, from those receipts). *Proposed*, the author's: that map is the reader's representation of the experience, and it is the knower's semi-norm of Section 4, the comparison read one way (*measured* as a semi-norm; *proposed* as what the experience is measured in). A frame fixes it: for a direction $u$ of interval one, $(u \cdot v)^2 - v \cdot v = 90\,(u_0 v_1 - u_1 v_0)^2$, never negative and blind exactly along $u$ (*measured*; *standard*, the rest-space metric $g + u \otimes u$). *Proposed*, the author's: the semi-norm is derived from the reference frame entirely; which frame is

the knower's. The comparison so built has a name, *same difference*: a controlled failure to confuse two measurements, nothing when read both ways, the frame's semi-norm when read one way (a *definition*, in the patter's words); the slip is its control (*proposed*, the author's).

> The device is a consuming iterator on the inverse of the refinement operator. Its stationary values are the roots, its derivatives are a geodesic and electrodynamics, and its polynomial, the quarter, is never seen. The reading is a projection with a residue.

# A Sources

This appendix alone names the middle card's declarations; all are in `Episode12.lean` unless marked.

**Section 1**

`every_table_satisfies_its_own_polynomial`, `the_table_of`, `the_crossings_are_a_quarter_turn_and_a_swap`, `both_roots_solve_any_table`, `the_walks_keep_a_three_term_recurrence`, `the_lanczos_recurrence_ends_at_the_table`; the lift and its lowering: `Episode02.lean` `Metavariable`, `Episode11.lean` `TYPESET`, `Episode10.lean` `REVIEWED`.

**Section 2**

`the_rayleigh_quotient_lies_between_the_roots`, `the_roots_are_reached_on_the_diagonals`, `the_residual_is_the_derivative_of_the_action`, `the_mirror_is_the_device_residual`.

**Section 3**

`Episode08.lean` `proportional_laws_accumulate_commutation_projections_called_TERM_LAMBDAS`, `the_device_computing_the_next_configuration`; `the_reading_with_its_receipt`, `the_walls_print_the_reading`, `the_lanczos_polynomial_is_the_residual`.

**Section 4**

`the_one_way_step_squared`, `the_semi_norm_is_the_comparison_read_one_way`, `read_both_ways_the_comparison_measures_nothing`, `the_floor_reads_the_step`, `both_ways_the_step_is_zero`, `the_semi_norm_is_never_negative`, `the_semi_norm_sees_nothing_exactly_on_the_doublers`, `the_still_and_the_alternating_are_its_kernel`, `on_the_turning_fields_the_derivative_is_twice_the_quarter_turn`, `the_galerkin_matrix_is_invertible`, `the_jacobian_free_difference_is_exact`, `one_jacobian_free_newton_step_stays_in_the_bracket`; `the_box_has_one_point`, `the_device_reads_it_back`; `Episode03.lean` margin on the implied norm.

**Section 5**

`gateauxAt`, `Episode04.lean` `Variation.the_gateaux`, `the_outer_and_the_hole`, `the_speed_is_below_light`.

**Section 6**

`frechetAt`, `Episode04.lean` `Variation.the_frechet`; `the_antisymmetric_part_is_the_quarter_turn`, `the_three_turns_square_to_minus_one`, `the_turns_compose_around_the_circle`, `every_torus_is_gauge_invariant`, `a_flat_torus_is_stationary`, `the_device_supplies_its_own_gauge`, `the_steps_commute`, `every_four_torus_is_gauge_invariant`, `a_flat_four_torus_is_stationary`, `the_gammas_anticommute`, `the_i_commutes_with_every_`

gamma, the_i_turns_the_metric_over, the_link_read_backward_turns_over, the_alternating_field_is_a_doubler, the_whole_action_is_gauge_invariant, the_whole_action_moves_only_with_$\beta$.

**Section 7**

theConvectiveGap.

**Section 8**

the_refinement_is_its_own_derivative, the_action_varies_by_the_operator_twice, the_surd_products_commute, the_refinement_commutes_with_every_boost, the_boost_preserves_the_interval, the_one_way_split_is_a_change_of_clocks, the_same_round_trip_is_the_same_clock, the_speed_is_read_from_the_multipliers, the_mixed_differences_commute, the_curl_of_a_gradient_is_nothing; Episode04.lean margin, "Einstein's Field Equations for $c$ a Lagrange multiplier"; Episode03.lean margin on bullshit, time, energy and mass, linearly related.

**Section 9**

the_four_labels_walk_perfectly, the_quarter_walk_repeats_every_four, the_other_phase_keeps_its_residue, the_other_phase_never_lands_on_a_label, the_norm_of_a_product, the_other_phase_stays_on_the_ring, the_panel_error_is_orthogonal_to_the_line, the_line_is_the_least_energy, refining_a_panel_never_lowers_the_line, the_line_squared_is_the_quartic; Episode03.lean Spline, and the margin "$\exists\,\delta\ \forall\,\varepsilon\ \ldots$ over and over"; Episode02.lean fenceposts.

**Section 12**

the_argument_counted_from_the_inferred_end, no_reader_counts_the_false_meaning_trues, the_click_table_is_derived, the_tally_of_the_numeral, the_roots_are_reached_on_the_diagonals, the_speed_is_read_from_the_multipliers, the_refinement_commutes_with_every_boost, the_antisymmetric_part_is_the_quarter_turn, two_crossings_turn_the_phase_over, the_other_phase_never_lands_on_a_label, every_four_torus_is_gauge_invariant, the_device_square_is_twice_wilson_less_its_square, the_i_turns_the_metric_over, the_dirac_action_is_gauge_invariant, the_curl_of_a_gradient_is_nothing, the_whole_action_is_gauge_invariant, the_refinement_is_its_own_derivative, the_readers_beta, the_readers_charge_is_bracketed, the_schwinger_term_from_the_reading, the_identity_does_not_click, the_quartic_deflates, the_deflated_cubic_factors, a_root_deflates_to_the_cubic, the_reading_solves_the_counted_quartic, the_conjugate_solves_the_counted_quartic, the_walls_print_the_reading, the_reader_adds_no_digit, outer_reader_agrees, the_other_machine_agrees, the_frame_gives_the_semi_norm, the_rest_length_is_never_negative, the_frame_cannot_see_itself, the_boost_is_a_frame.

**Sections 10–11**

the_hex_digits_print_pi, the_readers_beta, the_knowers_beta, the_readers_charge_is_bracketed, the_knowers_charge_is_bracketed, the_conjugate_solves_the_counted_quartic, the_two_roots_are_the_counts, the_quartic_sign_is_the_velocity, the_quartic_does_not_read_the_sign, every_boost_keeps_the_interval, the_hole_never_closes, the_ratio_of_the_roots_is_twice_the_boost, the_schwinger_term_from_the_reading.

## References

[1] Peter J. Mohr, David B. Newell, Barry N. Taylor, and Eite Tiesinga. CODATA recommended values of the fundamental physical constants: 2022. *Reviews of Modern Physics*, 97(2):025002, 2025.